FINANCIAL FORECASTING

DEFINITION OF RECURSIVE ALGORITHMS:

Recursion is the technique of making a function call itself. This technique provides a way to break complicated problems down into simpler problems which are easier to solve.

EXAMPLE: Addition of n numbers by using Recursion

```
public class Main {

  public static int sum(int k) {

    if (k > 0) {

      return k + sum(k - 1);

    } else {

      return 0;

    }

  }

  public static void main(String[] args) {

    int result = sum(5);

    System.out.println(result);

  }

}
```

Explanation:

Intialy k = 5

It enters if block 5>0 , it return 5 + sum(5-1)

for sum(4) it enters to the methods in if block 4>0 then it returns 4 + sum(4-1)

for sum(3) it enters to the methods in if block 3>0 then it returns 3 + sum(3-1)

for sum(2) it enters to the methods in if block 2>0 then it returns 2 + sum(2-1) for sum(1) it enters to the methods in if block 1>0 then it returns 1 + sum(1-1)

for sum(0) it enters to the methods in if block 0 = 0 then it returns 0

so sum(1) = 1 + 0 = 1

sum(2) = 2 + 1 = 3

sum(3) = 3 + 3 = 6

sum(4) = 4 + 6 = 10

sum(5) = 5 + 10 = 15

so it return 15 to main class calling area and prints 15

TIME COMPLEXITY:

In my Financial Forecasting solution, for every call creates an another call which takes O(n) time .

But if it extended to branching recursion like multiple investment then it becomes O(2^n) or worse.

Problem with simple recursion

- Recalculates same values again and again
- Very slow for large inputs

OPTIMIZATION TECHNIQUES:

To Optimize the recursion time complexity, store the already computed values in an array so that it takes lesser time than simple recursion. This method is call as Memoization.